

Distance Measures I

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Topics

- 2 Recap
- 3 Introduction
- 4 Eigenanalysis
- 5 Distance Measures
 - Desirable Properties of Distance Measures
- 6 Coming Up
- 7 Exercises and References

Recap

- Last time we covered **Introduction to Multivariate Statistics**:
 - A quick introduction to what we will do in this course.
 - Matrix multiplication.
 - Matrix transposition and symmetry.
 - Matrix inversion.

Introduction

- Distance measures are important for many forms of data analysis.
 - *In this course, it is very important for cluster analysis.*
- The distance measure selected needs to be done so based on the data you are working with.
 - *Garbage in garbage out.*
- First, we are going to discuss Eigenanalysis.
 - Very important for dimension reduction.

Introduction to Eigenanalysis

- **Eigenanalysis** is a method used to identify a set of linear equations that summarize a square matrix.
- The approach yields a set of **eigenvalues** (λ) which have an associated **eigenvector** ($\mathbf{X}$):

$$\mathbf{A}\mathbf{X} = \lambda\mathbf{X} \quad (1)$$

- The multiplication of a square matrix $\mathbf{A}$ and a vector $\mathbf{X}$ will yield the same values as the multiplication of a scalar value λ and the vector $\mathbf{X}$.
- This allows us to manipulate our individual data points in a coordinate-space by multiplying them by an eigenvector $\mathbf{X}$.
 - *Very important for PCA.*

Eigenanalysis

- Each dimension of a square matrix is represented by an eigenvector and associated eigenvalue.
- The dimensions are decreasing in order of importance.
 - Allowing us to focus on the first few dimensions and still see broad patterns in our data.
- If all eigenvectors are used, the patterns in the data are perfectly preserved.
 - *The dataset itself may still be manipulated.*

Spectral Decomposition

- In R we can use the `eigen()` and in Python `numpy.linalg.eig()` to obtain the eigenvalues and eigenvectors of a square matrix.
- The eigenvalues are **always** in order of decreasing size.
- Each eigenvector is associated with the eigenvalue in the same relative position.

Example 1

- Create the following matrix:
 - `A <- matrix(1:9, nrow = 3)`
- And do the following calculations:
 - 1 Calculate the eigenvalues and eigenvectors for this matrix.
 - 2 Use the code to verify Equation 1.
- What did you notice?

Trace and Determinant

- The **trace** of a matrix is equal to the sum of its diagonal values.
 - The trace is also the sum of its eigenvalues.
 - In R: `sum(diag(A))` or `sum(Eigenvalues)`
 - In Python: `numpy.trace(A)`
- The **determinant** of a matrix is equal to the product of its eigenvalues.
 - In R: `det(A)` or `prod(Eigenvalues)`
 - In Python: `numpy.linalg.det(A)`

Example 2

- Using your matrix from Example 1 do the following calculations:
 - ① Calculate the trace of $\mathbf{A}$ (both ways).
 - ② Calculate the determinant of $\mathbf{A}$ (both ways).
- What did you notice?

Singular Value Decomposition

- Another way to examine the eigenvalues and eigenvectors is through singular value decomposition.
 - In R: `A.svd <- svd(x = A)`
 - In Python: `numpy.linalg.svd(A)`

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T \quad (2)$$

- Outputs:
 - `A.svd$d` is analogous to `E$values`. Each of these is an **eigenvalue**.
 - `A.svd$u` is analogous to `E$vectors`. It is a set of vectors, each of which is associated with an eigenvalue and relates to the rows of the original matrix.
 - `A.svd$v` is another set of vectors. Each vector is associated with an eigenvalue, but relates to the columns of the original matrix.
- *We may look into this in more depth later on.*

Example 3

- Using your matrix from Example 1 compare the results of the `eigen()` function and the `svd()` function.
- What did you notice?

Eigenanalysis Comments

- We use these values to rotate, reflect, stretch, or compress data in a coordinate-space.
- Due to the fact that eigenvectors are in descending order, we can use this information for data reduction.
- *It is okay if you are a bit confused at this point.*

Distance Measures

Introduction to Distance Measures

- When we think about clustering, we want to group *close* observations together.
- How do we actually determine if observations or groups are close together or far apart?
- *We will cover some basic concepts and properties of distance measures.*

Illustrative Example 1A

- Assume that we have the following uni-variate dataset:

Plot	Plant Richness
J	4
K	10

- What is the *distance* between these two observations?

Illustrative Example 1 Cont.

- Next, we decompose the total plants by annual and perennial plants:

Plot	Annual	Perennial
J	3	1
K	7	3

- What is the *distance* between these two observations?

- Hint:

$$ED = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- What happened?

Desirable Properties of Distance Measures

1. Zero if Identical

- Very simply, if objects C and D have the same values for all variables, the distance between them should be **zero**.
- **This should be true of any distance measure we use.**
- Example:

Plot	Annual	Perennial
J	3	1
K	7	3
L	3	1

2. Positive Distance

- If objects C and D **do not** have the same values for all variables, the distance between them should be **greater than zero**.
 - *This should be true of any distance measure we use.*
- *What does a negative distance mean?*
- Example:
 - What is the Euclidean distance from J to K and from K to J in Illustrative Example 1?

3. Symmetric Distance

- A distance measure is **symmetric** if the distance from A to B equals the distance from B to A.
 - *This should be true of any distance measure we use.*
- Obviously, there are many real-life scenarios where this does not hold.
 - Travel times and distances, time-dependent changes in variable outcomes,...

4. Metric or Semimetric?

- We may calculate distances between many pairs of objects.
 - Assume we have three objects, this means that we can calculate three distances.
- A **metric** distance measure follows Euclidean geometry (triangle inequality theorem).
- Measures that **do not** follow the triangle inequality theorem are **semimetric**.
 - It is possible for this to happen and these measures are called **dissimilarity measures**.

5. Constant Maximum?

- Some measures do not have an upper limit.
 - *No matter how physically far apart two points are, you can envision two points that are further apart.*
- Other measures have a constant maximum - *no scenario where a distance could be greater.*
 - Typically happens when objects have no elements in common.
 - Usually occurs when distance is expressed as a proportion of some total (dissimilarity measure).
 - Often allows for a measure of similarity.

Example 4

- Assume that the following objects all have binary attributes.

Student	D.S. Student?	Undergrad Student?	Off Campus?
A	0	1	0
B	1	1	1
C	1	0	1

- What measures of distance/dissimilarity could we use?
- Will any of these measures have an upper limit?

The Distance Matrix

- The pairwise distances between objects can be represented as a **distance (dissimilarity) matrix**.
- Properties:
 - ① The distance matrix is square - *useful for matrix manipulations*.
 - ② It is symmetric (property 3).
 - ③ All values along the diagonal are zero.
 - ④ The first row and last column are non-informative (symmetric distance).
 - *Can display using a lower triangular matrix*
- The number of unique pairwise combinations: $\frac{n(n-1)}{2}$.

Example 5

- Use Euclidean distance to obtain the distance matrix for these three observations:

Plot	Annual	Perennial
J	3	1
K	7	3
M	4	2

- What do you notice?

Comments on Distance Measures

- Calculations of a distance between two objects compresses all of the differences into a single value.
- Distance-based techniques can be applied identically to both univariate and multivariate data.
 - *Function of the number of objects and not the number of variables.*
- We may use a distance matrix to test for differences among pre-existing groups, visualize patterns, or identify natural groups in the data.

Road Map

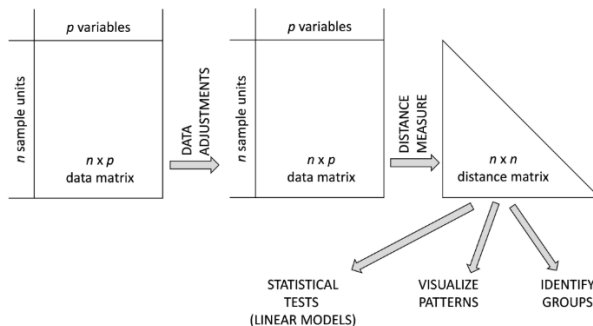


Figure: source: (1)

Next Lecture

- We will discuss more **Distance measures**.
- Common Distance Measures.
- *Multivariate Outlier Analysis*

Exercise 1

- Use Euclidean distance to obtain the distance matrix for these four observations:

Plot	Annual	Perennial
J	3	1
K	7	3
M	4	2
P	3	3

- What do you notice?

Exercise 2

- Using your distance matrix from Exercise 1:
 - 1 Calculate the eigenvalues and eigenvectors of this matrix.
 - 2 Calculate the trace of the matrix.
 - 3 Calculate the determinant of the matrix.
 - 4 Perform Singular Value Decomposition of the matrix.
- What do you notice?

References & Resources

- ① Bakker, J. D. (2026) Applied Multivariate Statistics in R
 - ② Peng R. D. (2020) *Exploratory Data Analysis with R*
 - ③ Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction*. Chapman and Hall/CRC
- `svd()`